

Mani Sathish Pabolu

Design Engineer -
Integration



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🎓 EDUCATION

Electrical Engineering

AMIE

05/2017 – present | Kolkata, India

Diploma in Electrical and Electronics Engineering

Sri Sai Aditya Instit of Sci & Tech

06/2013 – 04/2016

Surampalem, India

📜 CERTIFICATES

IPC 610-G and IPC 620-C

Expert

IPC J STD 001 & ISRO PAX 300

Expert

Mission planner

Expert

Manisathish119@gmail.com

👤 PROFILE

Skilled **UAV/UAS engineer** with over **9+ years** of experience in drone systems **fabrication, avionics integration, material management, and comprehensive functional and hardware testing**. Proficient in **software simulation, troubleshooting, flight log analysis, Autopilot firmware and parameters, propulsion system, and power distribution systems**. Experienced in **copter tuning, mission planning for automated and manual flights, payload operation, and pre flight checks and flight readiness**. Adept at supervising production teams to ensure quality and efficiency.

Possesses 3 years of specialized expertise in wire harness and PCBA preparation for large-scale aircraft and military projects with clients such as **ISRO, Boeing, and Lockheed Martin**. **Certified in IPC J-STD-001, ISROPAX 300, IPC 610-G, and IPC 620-C** soldering and wire harness standards, with strong knowledge of **AS9100, ISO 9001, Kaizen, and EMS 14001** quality management systems.

👛 PROFESSIONAL EXPERIENCE

UAV / Drone Integration & Testing Engineer

Sanad Academy & Streit Security Vehicles (R&D Division - SKYSTREAM)

11/2022 – 02/2026 | Abu Dhabi / Ras Al Khaimah, UAE

- Led **integration, testing, and troubleshooting** of UAV/UAS avionics, communication systems, and flight controllers including **Pixhawk, CUAV, NORA** and firmware **Ardupilot, PX4** and **Betaflight**.
- Conducted **pre-flight inspections, full functional checks, and safety validations** for VTOL, quadcopter, and fixed-wing platforms, ensuring **flight readiness and operational safety**.
- Planned and executed **autonomous mission flights**, performing **parameter tuning, flight performance optimization, and post-flight data analysis** for both small-scale and large-scale UAVs.
- Supervised **wiring harness assembly and avionics installation**, ensuring **proper connections, signal integrity, and system reliability**.
- Designed, modeled, and **3D-printed new UAV subsystem components** using **SolidWorks, Creo, and AutoCAD**, contributing to R&D development of prototype UAVs.
- Validated **system performance** through rigorous **SIL testing** and led comprehensive **end-user training** to ensure seamless software adoption and operational reliability.

SKILLS

UAV / Systems ● ● ● ● ●

Engineering:

- UAV Electrical Architecture & Avionics Integration, Ardupilot, PX4
- Mission Planning, Payload Operations
- Firmware, Parameter and End to End Testing of UAV/UAS

Integration of UAV Systems ● ● ● ● ●

- Frame Assembly
- Wiring Preparation
- Harness Routing/Installation

Functional Testing of UAV ● ● ● ● ●

- Validating the Propulsion System, PIDs and Electronics
- Payload Systems and Sensor Integration

Avionics Readiness ● ● ● ● ●

Firmware for Autopilot, ESCs, Motors and Testing Regulators, Servos, Radios, Pixhawk,

Electrical & Panel Design: ● ● ● ● ●

- Circuit Schematics & Wiring Diagrams
- Harness Design, Routing & Termination

Software & Design Tools: ● ● ● ● ●

- AutoCAD (Schematics & Harness Layouts)
- Creo Draft 2D & Creo Parametric 3D
- 3D Modelling & Rapid Prototyping (SolidWorks / 3D Printing)

PCB Assembly ● ● ● ● ●

Soldering, Chemical Bonding, Mechanical fixing

- Engineered a complete **UAV Ground Control Station** from scratch, designing the full **electrical architecture** and **power systems**.
- Developed **flight manuals, checklists, standard operating procedures, and mission plans**, ensuring **consistent operational standards** across projects.
- Managed **firmware loading, programming, and verification of UAV systems**, including radios, sensors, and actuators to maintain **high-performance operations**.
- Coordinated with **international clients and stakeholders**, delivering **theoretical and practical UAV training**, demonstrating technology capabilities, and supporting **custom UAV solutions**.
- Oversaw **inventory management and procurement of UAV parts**, maintaining **cost efficiency and quality standards**.

Key Achievements:

- Successfully **planned and executed light drone shows (AIR GLOW)** that won the **Guinness World Record for two consecutive years**, demonstrating precision in software/hardware coordination and team leadership.
- Enhanced UAV system reliability through **rigorous testing and post-flight analysis**, leading to **improved mission success rates and reduced system failures**.
- Contributed to **prototype UAV design and subsystem innovation**, improving operational efficiency and R&D capability within the organization.

Electrical Design Engineer @ Applied Materials

HCL Technologies Ltd

07/2022 – 11/2022 | BANGALORE, India

- Designed **line diagrams, 3D cabling layouts, wiring harnesses, and 2D drawings** for prototype and production using **AutoCAD and CREO Parametric**.
- Prepared **schematics for power, signal, and grounding harnesses** using **CREO Schematics**.
- Managed **Engineering Change Orders (ECO/ECN)**, updating **drawings, BOMs, and specifications** through **Windchill**.
- Selected and integrated **switchgear components** (MCBs, fuses, relays, contactors, isolators, power supplies) based on **load and protection requirements**.
- Designed **panel-level protection circuits** for overcurrent, short-circuit, and isolation to enhance **system safety**.
- Worked on **CREO modelling and assembly** to support prototype and production readiness.

PROJECTS

Unmanned Aerial Vehicle and Systems

Avionics Integration | Functional Testing | Flight Operations

2018 – present

Executed avionics integration, functional testing, and flight operations for UAVs from 2 kg FPV drones to 150 kg heavy-lift systems, configuring flight controllers, telemetry, navigation, FPV, and gimbal cameras for surveillance, loiter, reconnaissance, and special operations in defense and non-defense sectors.

- **Heavy-lift multirotors:** 90 kg & 150 kg octa-quadcopters for long-endurance loiter and surveillance missions.
- **VTOL UAVs:** 20 kg, 45 kg, and 50 kg platforms with 2–10 kg payloads, ranges up to 100 km, and gimbal/FPV integration.
- **Payload Quadcopters:** 18 kg & 75 kg systems carrying 5–40 kg payloads for varied mission profiles.
- **FPV drones:** 2 kg & 6 kg high-speed kamikaze platforms with real-time video and telemetry systems.
- **Fixed-wing UAV:** 8 kg surveillance platform with integrated 3-axis gimbal camera.

LANGUAGES

English	● ● ● ● ●
Telugu	● ● ● ● ●
Kannada	● ● ● ● ●
Hindi	● ● ● ● ●
Tamil	● ● ● ● ●

Integration Engineer @ Small UAV Lab

Tata Advanced Systems Ltd

08/2019 – 07/2022 | BANGALORE, India

- Designed and prepared **wiring harnesses for prototype UAVs (VTOL, quadcopters, fixed-wing)** using **Creo Parametric 3D** and **AutoCAD 2D schematics**, including **BOM preparation**.
- Executed **end-to-end wiring** for avionics, autopilot, PSU, GPS, gimbal, RF cables, and control systems.
- Hands-on experience with **manual soldering, harness assembly, avionics integration, and battery maintenance**.
- Tested and validated **avionics, radios, and aircraft electrical systems** prior to installation.
- Performed **pre-flight checks, log maintenance, troubleshooting, and corrective actions** to meet project timelines.
- Participated in demonstrations for **CRPF, BSF, Indian Army, and Navy expos**.
- Documented assembly, testing, and lab activities, ensuring adherence to **lab protocols, process documentation, and quality standards**.
- Supported **mechanical integration, daily monitoring, and coordination with cross-functional teams**.

Assistant Engineer for Wire Harness

Rossell Techsys

10/2018 – 08/2019 | BANGALORE, India

- Guided team members in achieving **monthly production and shipment targets**, ensuring efficiency and quality.
- Conducted **soldering in critical areas** and provided **hands-on training** to team members.
- Applied expertise in **connectors, cables, and electrical materials**, ensuring correct selection and handling.
- Maintained **lab and work environment**, adhering to safety and cleanliness standards.
- Performed **stripping, crimping, insertion, terminations, and cable laying**, including **metal and nylon braid terminations**.
- Managed **chemical handling** for processes involving CC, resins, Araldite, and potting compounds.
- Manufactured and tested **high-voltage cables**, troubleshooting issues and ensuring compliance with specifications.
- Conducted **in-process inspections** at various stages and performed **final visual inspections** with comprehensive **audit report preparation**.



AWARDS

Star of the Month and Best team Award

Tata Advanced Systems

05/2020

Achieved Multiple Awards for the UAV systems prototyping

Best Improvement Award

Centum Electronics Pvt Ltd

10/2017

Achieved best improvement award for making a Harness Checking Zig

2nd Place at Model Making

Inter Colleges tournament

VEDAS

01/2015

Achieved 2nd place in Volcano Power generation model Making design

1st place for District level chess competition

District level school board

09/2011

Achieved 1st place in District level Chess competition in Junior Level category

Senior Technician for PCB Assembly

Centum Electronics Pvt Ltd

04/2016 – 10/2018 | BANGALORE, India

- Performed **soldering of various components** and managed **chemical handling** with proper waste records.
- Conducted **final visual inspections** and prepared **final audit reports** to ensure product quality.
- Executed **internal audits** in compliance with **AS 9100, ISO 9001, EMS 14001, and IMS** standards.
- Trained team members in **4M, 5S, 6 Sigma, and 7 QC tools** to improve operational efficiency.
- Maintained **equipment calibration records** and ensured compliance with process requirements.
- Tested **assembled units**, performed **troubleshooting**, and implemented corrective actions.
- Managed **inventory control, material planning, and manufacturing resource planning**.
- Conducted **incoming goods inspection** for raw materials and components, ensuring compliance with specifications.



DECLARATION

I hereby declare that the above information provided by me is true to the best of my knowledge and belief and to work hardheartedly for the organization.

Mani Sathish Pabolu